Boston 311 Crew Staffing: Analysis Report

IE 5374 Applied Generative AI Hackathon, Topic 2: Staffing Boston's 311 City Service Teams

Team: Jogeashwini Srinivasan Ramesh, Kavin Priyadarrsan Murugesan and Chitra Pahuja

1. Summary

Our main recommendation is to grow Transportation from 18 crews to 25 over 2025. At 18 crews it falls further behind every month and only 7% of its requests close on time; with the extra crews that rises to 93%, and the department's cost drops from $10.2M to $2.8M because it stops paying late fines. The other three departments need only small changes. The full plan costs about $13.1M.

We built the plan for Public Works (PWDx), Transportation (BTDT), Parks and Recreation (PARK) and Inspectional Services (ISD) from the 311 requests the city received in 2023 and 2024. We also built an intake assistant that reads a resident's message and works out the request type, department and queue. On our test messages it had the right type in its top three every time a message was clear.

The plan doesn't leave much slack. If demand rose 20%, or a bad snowstorm hit, Transportation would be overwhelmed again, so we suggest carrying one or two extra Transportation crews into the winter.

2. Data and forecast

We used the city's 311 data for 2023 and 2024 and kept only the four departments, which left 558,663 requests. We didn't use 2025 data because the city switched 311 systems in October 2025. The code is in `src/data.py` and `src/forecast.py`.

Our 2024 numbers match the brief exactly: 157,610 requests for Public Works, 81,447 for Transportation, 18,736 for Parks and 18,624 for Inspectional Services, with 90%, 50%, 93% and 72% closed on time. The on-time figures only match if every request is counted. About 11% of 2024 requests have no target date, and the data marks all of them on time; dropping them would take Public Works down to 87%. We count them, as the brief does, and we count every request as work no matter how it was closed.

Demand follows the seasons. Public Works is busiest in August and September, and Parks more than triples between January and June. Public Works also shrank by 8% from 2023 to 2024, while the other three departments grew by 3 to 8%.

Two years gives only 24 monthly points per department, which is too few to fit a proper time-series model. So we kept it simple: for each month we took its average share of the year, and multiplied that by a 2025 total that continues the 2023-to-2024 trend for one more year. To see how well this works, we used 2023 alone to predict 2024 and measured the average monthly error.

Department	2023	2024	Change	2025 forecast	Backtest error
Public Works	171,268	157,610	−8.0%	145,040	13.6%

Department	2023	2024	Change	2025 forecast	Backtest error
Transportation	75,314	81,447	+8.1%	88,078	7.6%
Parks and Recreation	17,541	18,736	+6.8%	20,010	17.7%
Inspectional Services	18,123	18,624	+2.8%	19,140	13.3%

Transportation is the easiest department to predict. Parks is the hardest, mostly because its 2024 peak came in June rather than August. Most of the Public Works error comes from its 8% drop, and since the forecast assumes that drop continues into 2025, it is the number we would keep the closest eye on. The charts are in `docs/figures/monthly_demand.png` and `docs/figures/forecast_2025.png`.

3. Intake assistant

The intake assistant takes a message written in a resident's own words and returns the request type, the department and the queue. When the message is too vague to route, it asks the resident one follow-up question instead. The code is in `src/catalog.py`, `src/intake.py` and `app/app.py`.

It starts from a catalog of the 158 request types the four departments handled in 2024. For each type we stored the department and queue that handled it most often and a one-sentence description in everyday language, which gpt-4o-mini wrote and we checked. When a message comes in, text-embedding-3-small finds the eight types whose descriptions are closest to it, and gpt-4o-mini picks one of them. The model has to answer in a fixed format: the type, two alternatives, how confident it is, and whether a follow-up question is needed. It only chooses the type. Our code then looks up the department and queue in the catalog, so the model can't invent either one. If it ever names a type that wasn't on the shortlist, we ask again once and then flag the answer.

Before running any model we wrote 44 test messages the way residents actually write, typos included. 39 have a clear answer (17 for Public Works, 10 for Transportation, 6 each for Parks and Inspectional Services, and two that report two problems at once). The other 5 are deliberately vague, so the right response is a question. We compared three versions of the prompt:

Version	Department	Type, top 1	Type, top 3	Queue	Asks when unclear	Quiet when clear
V1: all 158 type names, no examples	97%	85%	90%	79%	20%	100%
V2: V1 + 6 worked examples	97%	90%	92%	82%	80%	100%
V3: top 8 retrieved types + examples	95%	90%	100%	77%	100%	97%

With retrieval, the correct type was always among the eight candidates. Adding worked examples mainly changed when the model asks for more information: on vague messages it went from asking 20%

of the time to 80%. V3 had the right type in its top three for every clear message and asked a question on all five vague ones, so that is the version in the app. Queue accuracy is lower than department accuracy because many queues are city districts, and a message usually doesn't say which district it's in.

V3 got four request types wrong, and all four were close calls between similar types. Three of them came down to the catalog descriptions rather than the prompt. "Valet Parking Problems", with 7 requests in 2024, beat "Parking Enforcement", with 63,064, because its description happened to mention a blocked driveway. In the same way, Parks' "Graffiti Removal" (12 requests) beat Public Works' "PWD Graffiti" (1,036). Better descriptions, or showing the model how common each type is, would fix most of these. Two caveats: 44 messages is a small test, and the same AI tool helped write both the test messages and the descriptions, so real residents would probably see somewhat lower accuracy. The full error analysis is in `docs/intake_error_analysis.md`.

4. Staffing model

To plan crews we wrote a simulator that plays out a department's year one day at a time, and a planner that searches for the cheapest month-by-month plan by running each candidate through the simulator. The code is in `src/simulator.py` and `src/planner.py`, and every number comes from `config.yaml`.

In the simulator, each month's forecast is spread evenly across its days, and every request gets a due date drawn from its department's 2024 mix of targets. Crews work weekdays only. A month's capacity is the number of crews times the requests a crew closes per day times 21 working days, and crews always work on whatever is due soonest.

Each crew costs a twelfth of its annual pay every month. A request that is still open after its target costs $20, and one still open 30 days after its target costs another $100.

The planner starts from four different plans: January's crews kept all year, a plan that follows demand, and two in between. From each one it tries every small change, such as one more or one fewer crew in a single month, or from a given month through December, plus seasonal crews for Parks. It keeps whichever change saves the most and repeats until nothing helps. January stays at the starting crews, because any change needs a month of hiring or notice first.

From the brief: a new crew trains for a month before it starts working; releasing a crew takes a month's notice and costs 25% of a year's pay; and Parks can add seasonal crews from April to October at 1.3 times the cost, with no training needed.

Where the brief doesn't say, we made these choices:

- Crews are paid while they train and during their notice month.
- Requests with no target date get a 365-day target, so they count as on time, which is how the data treats them.
- Any request still piled up on December 31 costs $120, apart from the last 14 days of arrivals that haven't had time to be closed. Without this, a plan could look cheaper just by pushing work into the next year.
- A seasonal crew closes as many requests a day as a full-time Parks crew (6).

5. Check against 2024

To see how realistic the simulator is, we ran 2024's actual demand through it with the January 2025 crew numbers kept all year and compared the result with what really happened.

Department	Crews	Requests, 2024	Simulated on time	Actual on time, 2024
Public Works	50	157,610	96.3%	89.7%
Transportation	18	81,447	17.3%	49.9%
Parks and Recreation	10	18,736	99.3%	93.2%
Inspectional Services	16	18,624	99.5%	72.4%

For Public Works, Parks and Inspectional Services the simulator is too optimistic, because it assumes crews spend every available hour on the most urgent request. Real crews don't work that way. Transportation goes the other way. About 20% of its 2024 requests were never closed, which suggests its crews kept up with new requests and let some older ones go. The simulator works through every request in order, falls behind, and ends up with almost everything late.

The brief asks us to use its crew capacities, so we didn't tune the model to close this gap. We report it instead, and read the plan as the minimum staffing needed at the brief's capacities, not as a forecast of the service residents will get.

6. The 2025 staffing plan

Transportation needs the biggest change. Going from 18 to 25 crews raises its on-time rate from 7% to 93% and cuts its 2025 cost from $10.2M to $2.8M. Across all four departments the base plan costs about $13.1M in crew pay and fines.

Department	Crews by month	Hires / releases	Keep January's crews: on time, cost	Plan: on time, cost
Transportation	18 in January, 22 in February, rising to 25 by September	7 / 0	7.3%, $10.21M	93.0%, $2.77M
Public Works	50 in January, 48 from February	0 / 2	96.4%, $6.60M	96.4%, $6.43M
Parks and Recreation	10 until March, 12 in April, 15 from May	5 / 0	99.2%, $1.84M	99.4%, $1.67M
Inspectional Services	16 all year	0 / 0	99.4%, $2.53M	99.4%, $2.53M

The costs in this table include the year-end charge from Section 4. The app and `staffing_plan_2025.csv` show only the crew pay and fines within 2025.

Keeping Transportation at 18 crews turns out to be the most expensive option. Those crews close fewer requests than come in, so the backlog grows all year and nearly everything ends up late. The fines add up to far more than seven extra crews would cost.

Public Works can let two crews go, because the forecast has its demand falling about 8%, as it did from 2023 to 2024. Inspectional Services can stay as it is.

We expected Parks to rely on seasonal crews, but the planner chose full-time crews instead. About 59% of Parks requests have a 365-day target, so work keeps arriving after October and crews are still needed in November and December. This result depends on the year-end charge we chose.

7. Sensitivity

We ran every scenario twice: once keeping the base plan as it is, and once letting the planner re-optimize for that scenario. The gap between the two shows how much it matters to plan ahead. The code is in `src/sensitivity.py`.

To size the spikes we looked for the worst months in our data. Both winters were mild: the worst winter month was only 7% above normal, and the worst single week 33% (Public Works). So our snowstorm is an assumed severe storm, adding 50% to all of February for Public Works and Transportation. The heat wave adds 40% to all of July for Parks and Inspectional Services, which matches the worst summer week we found (Inspectional Services, June 2023).

Scenario	**Department**	**On time, base plan kept**	**On time, re-optimized**	**Average crews vs base**	**Cost kept → re-optimized**
Demand +20%	Transportation	2.4%	86.1%	+5.1	$11.04M → $3.50M
Demand +20%	Public Works	56.6%	96.3%	+8.4	$10.37M → $7.47M
Demand +20%	Inspectional Services	51.9%	99.4%	+2.9	$3.48M → $3.06M
Demand +20%	Parks and Recreation	99.3%	99.4%	+2.5	$2.08M → $1.98M
Demand −20%	Public Works	96.5%	96.1%	−9.5	$6.41M → $5.60M
Demand −20%	Transportation	94.8%	93.6%	−4.8	$2.72M → $2.21M
Demand −20%	Parks and Recreation	99.4%	99.4%	−2.4	$1.67M → $1.38M
Demand −20%	Inspectional Services	99.5%	99.5%	−2.0	$2.24M → $2.15M
Snowstorm	Transportation	8.6%	93.0%	+1.4	$4.27M → $3.06M
Snowstorm	Public Works	96.3%	96.4%	+0.2	$6.49M → $6.45M

Scenario	Department	On time, base plan kept	On time, re-optimized	Average crews vs base	Cost kept → re-optimized
Heat wave	Inspectional Services	83.7%	99.4%	+0.4	$2.67M → $2.64M
Heat wave	Parks and Recreation	99.4%	99.4%	+0.7	$1.80M → $1.77M

Transportation is where things break. With 20% more demand its on-time rate falls to 2%, and after the snowstorm to 9%, with a backlog it still hasn't cleared by December 31. If the storm is planned for in advance, it only takes about 1.4 extra crews on average. Those crews have to be hired before the storm, though, because a new crew spends its first month in training.

Parks is the opposite case: its on-time rate looks fine, but its backlog doesn't. After the heat wave it still closes 99% of requests on time, yet the pile of open requests isn't gone by December 31, because most of its requests have year-long targets.

Higher service targets are cheap up to a point. The cheapest plan already beats 90% on time in every department, so targets up to 90% cost nothing extra. Transportation can't reach 95% at any cost; its best is about 94%, because crews only work weekdays and many of its requests are due the next day. See `docs/figures/cost_vs_service.png`.

8. Other considerations

We also looked at who gets served well, using 2024 data. The code is in `src/equity.py`. On-time rates count every request, and median days to close only counts requests that were closed.

Neighborhoods see the biggest differences in Transportation. Only 37 to 39% of requests in West Roxbury, Jamaica Plain and Hyde Park closed on time, compared with 62% in South Boston, and residents in those outer neighborhoods waited longest, a median of 3 to 4 days against less than a day downtown. Public Works is more even, between 80% and 93% on time, with Charlestown at the bottom. In Inspectional Services, Greater Mattapan and East Boston are lowest at about 64%. When Transportation adds crews, the neighborhoods that wait longest should be first in line. See `docs/figures/neighborhood_on_time.png`.

A lot of 311 requests don't come from residents at all. City employees log 62% of Transportation requests and 60% of Parks requests themselves, so a big share of crew workload is problems staff spot on their own.

Department	Logged by employees	On time, employee	On time, resident
Transportation	62%	50%	50%
Parks and Recreation	60%	95%	90%
Public Works	50%	90%	90%

Department	Logged by employees	On time, employee	On time, resident
Inspectional Services	40%	75%	71%

Resident requests are closed on time a little less often than employee requests in Parks and Inspectional Services. They also take longer to close everywhere except Parks, where they are actually faster. In Public Works the two on-time rates are equal, but that may hide a gap: 32% of resident requests there have no target date, and so count as on time, compared with 5% of employee requests.

9. Recommendations and limitations

In short, staff up Transportation first and give it a buffer for winter. The other departments need only modest changes.

1. **Hire Transportation crews now.** Four should start in February, building to 25 crews by September. This is the biggest improvement in service in the whole plan, and it saves money.
2. **Carry one or two extra Transportation crews into the winter.** A storm can't be handled by hiring once it starts, and the extra crews cost much less than the fines they prevent.
3. **Release two Public Works crews only if demand keeps falling.** Check after the first quarter.
4. **Grow Parks from 10 to 15 crews for the season,** and track its open requests as well as its on-time rate.
5. **Keep Inspectional Services at 16 crews.**

Limitations.

- The 2024 check in Section 5 shows a gap between the simulator and reality, so the plan is a minimum at the brief's capacities, not a forecast of service.
- The simulator spreads each month's demand evenly over its days and assumes crews always take the most urgent request.
- The year-end charge is our own assumption, and the choice of full-time over seasonal crews for Parks depends on it.
- The forecast gives one number per month. The ±20% scenarios are how we account for uncertainty in it.

10. AI use

The intake assistant is built on OpenAI models. gpt-4o-mini wrote the request-type descriptions and picks the request type for each message, and text-embedding-3-small handles retrieval. Their outputs are saved in `data/cache/`, so every result in this report can be reproduced.